KINETIC MODELING AND SIMULATION OF PLUG FLOW REACTOR FOR CATALYTIC REFORMING OF NAPHTHA USING MATLAB

1. Abstract:

This project models and simulates an industrial packed-bed Plug Flow Reactor (PBR) for catalytic naphtha reforming using **MATLAB.** To simplify naphtha's complex chemistry, we applied the Smith 4 - Lump kinetic model tracking Paraffins (P), Naphthene (N), Aromatics (A), and Gases (G), and solved the highly stiff mass and energy balance ODEs using the ode15s solver. The simulation captures a sharp endothermic temperature drop from 500°C down to 300°C, showing exactly how the reaction "freezes" early in an adiabatic bed due to a lack of thermal energy. Despite this chill, the net aromatic yield successfully reaches 90% relative to the initial naphthene feed, boosted by the conversion of paraffins. Ultimately, this model clearly demonstrates why real-world refineries cannot rely on a single reactor bed and must use multi-bed configurations with interstage heaters to maximize high-octane fuel production.

2. Introduction:

Catalytic reforming is a foundational process in modern petroleum refining, primarily utilized to upgrade low-octane naphtha into high-octane liquid stocks known as reformates (rich in aromatic hydrocarbons). These reformates serve as critical blending stocks for premium motor gasoline and provide chemical feedstocks like Benzene, Toluene, and Xylene (BTX). The primary target reactions involve the conversion of paraffins and cycloalkanes (naphthene) into aromatics. Because these reactions are carried out continuously at high temperatures (480°C to 520°C) and pressure (15 to 35 bar) over a platinum-based catalyst, industrial operations utilize fixed-bed Plug Flow Reactors (PFRs). This project mathematically models an adiabatic PFR to analyze how the composition, temperature, conversion, and yield vary across the catalyst bed weight (W).

3. Methodology:

Reaction Mechanism & Kinetics:

Naphtha contains hundreds of distinct hydrocarbon types, making individual molecular tracking computationally restrictive. Thus, the classic Smith (1959) 4-Lump Kinetic Model is implemented. The system tracks 5 main chemical species flowing through the reactor:

1. **Paraffins (P)**
2. **Naphthene (N)**
3. **Aromatics (A)**
4. **Light Gases (G)**
5. **Hydrogen $(H)_2$**

The three primary reactions considered are:

Reaction 1: Dehydrogenation of Naphthene to Aromatics (Highly Endothermic, Reversible)

$$\mathbf{N \rightleftharpoons A + 3H_2}$$

Reaction 2: Isomerization of Paraffins to Naphthene (Mildly Endothermic, Reversible)

$$\mathbf{P \rightleftharpoons N}$$

Reaction 3: Hydrocracking of Paraffins to Light Gases (Exothermic, Irreversible)

$$\mathbf{P + H_2 \rightarrow G}$$

Each reaction follows first-order kinetics with respect to the limiting reactant and is scaled by catalyst activity:

$r_1 = k_1 \left(P_N - P_A \bullet P^3_{H2} / K_{Eq.}\right)$

$r_2 = k_2 \left(P_P - P_N\right)$

$r_3 = k_3 \cdot P_P \cdot P_H$

4. Mathematical Modeling & Governing ODEs:

A steady-state elemental mass and energy balance across an incremental catalyst weight of dW yields the following set of ordinary differential equations (ODEs).

(i). Component Mass Balance:

$$\frac{dFp}{dW} = -r_2 - r_3$$

$$\frac{dFn}{dW} = r_2 - r_1$$

$$\frac{dFa}{dW} = r_1$$

$$\frac{dFg}{dW} = 2r_3$$

$$\frac{dFh2}{dW} = 3r_1 - r_3$$

(ii). Energy Balance:

For an adiabatic system where no heat is exchanged across the reactor walls, the change in temperature (T, in Kelvin) with respect to catalyst weight is driven strictly by the enthalpies of reactions ($\Delta H_{Rxn,j}$) and the collective heat capacities ($C_{P,i}$) of the moving fluid.

$$\frac{dT}{dW} = \frac{(-\Delta Hrxn,1 \cdot r1) + (-\Delta Hrxn,2 \cdot r2) + (-\Delta Hrxn,3 \cdot r3)}{\Sigma(Fi \cdot Cp,i)}$$

5. Simulation Methodology & Initial Conditions:

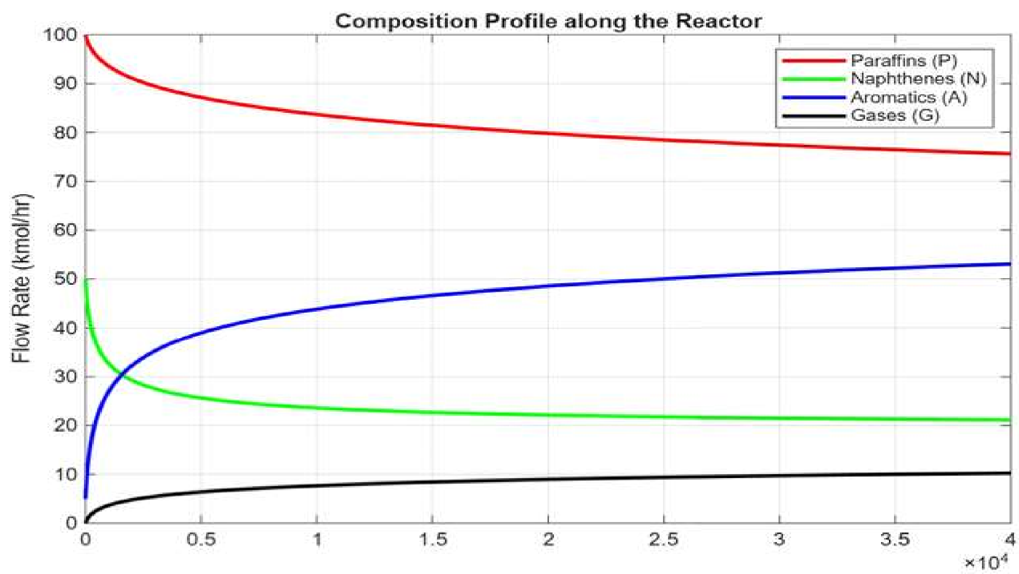
The model parameters, activation energies, and thermodynamic constants were tuned based on standard industrial baselines, and ode15s were used to integrate the 6 coupled ODEs over a catalyst weight range of 0 to 40,000 kg.

Parameter / Property	Unit
Operating Pressure (P_{total})	30 bars
Catalyst Weight Range (W)	0 to 40,000 kg
Initial Temperature (T_0)	500°C (K)
Initial Paraffins Flow (F_{P0})	100 kmol/hr
Initial Naphthenes Flow (F_{N0})	50 kmol/hr
Initial Aromatics Flow (F_{A0})	5 kmol/hr
Initial Hydrogen Flow (F_{H2O})	450 kmol/hr
Heat of Reaction 1 (ΔH_1)	210,000 kJ/kmol
Heat of Reaction 3 (ΔH_3)	− 45000 kJ/kmol

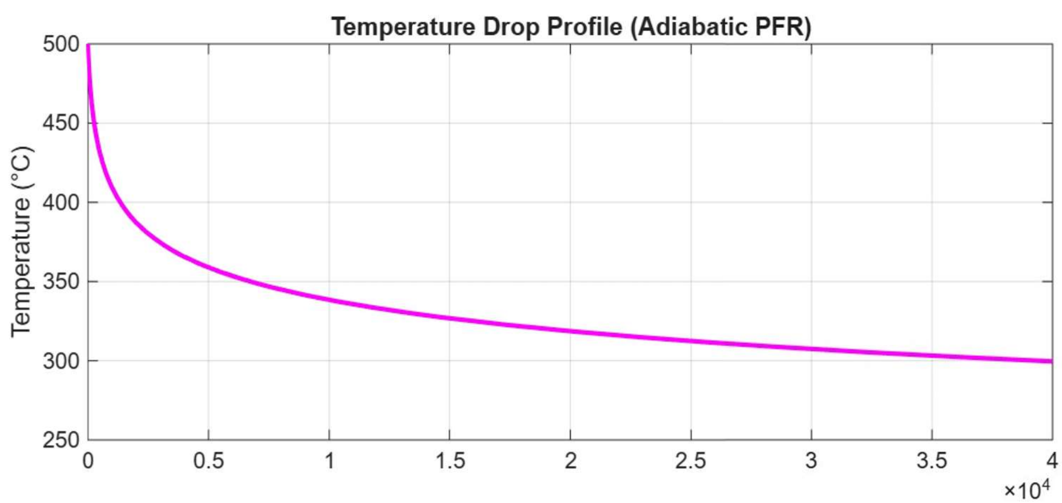
Because forward dehydrogenation operates at timescales, orders of magnitude faster than hydrocracking; the differential equations exhibit high numerical stiffness. MATLAB's **ode15s** (Variable-order standard stiff solver) was utilized rather than **ode45** to prevent computational divergence and ensure smooth, stable step-size profiles.

6. Results and Discussion:

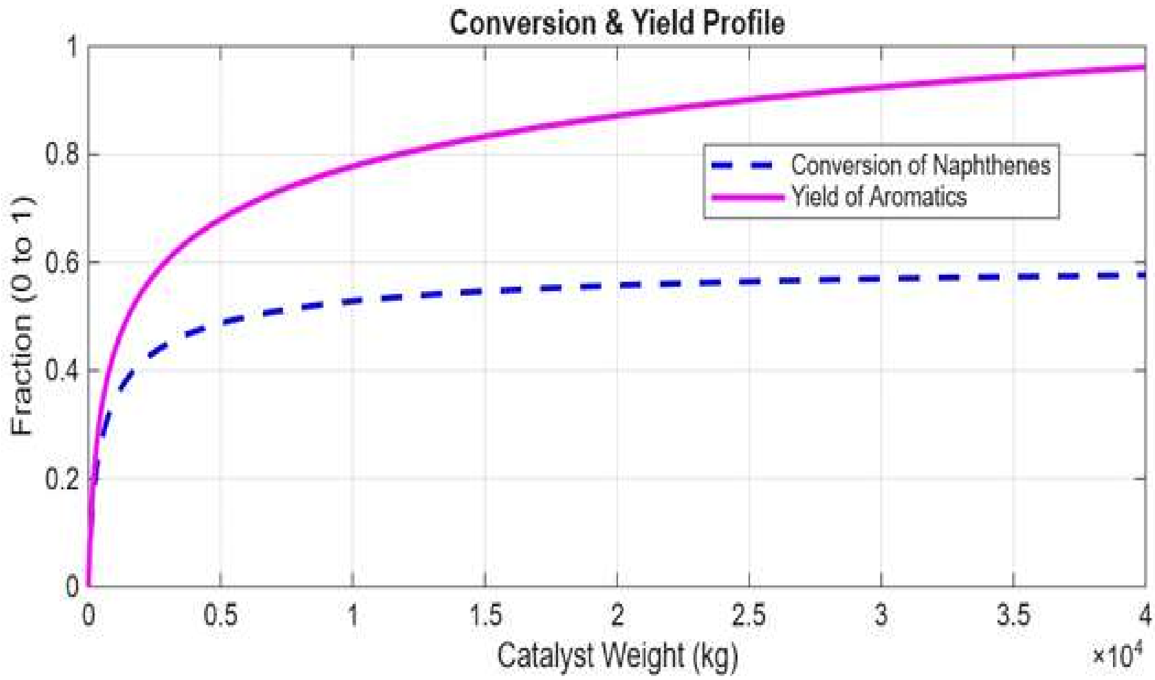
The MATLAB simulation produced the following outputs:



Paraffin decreases significantly due to conversion into Napthenes and Aromatic formation increases downstream due to dehydrogenation.



The reactor shows a steep temperature drop due to endothermic reactions, indicating heat consumption by dehydrogenation and cracking.



As seen in the composition profile, the concentration of Naphthenes drops rapidly within the first 15000 kg of catalyst and the remaining 25,000 kg of catalyst bed practically unutilized. Concurrently, the Aromatics profile rises from its initial 5 kmol/hr line up to a stable maximum of 53 kmol/hr. Paraffins consume gradually, while the undesirable cracked gas stream remains successfully restricted below 10 kmol/hr, proving high catalytic selectivity towards aromatic upgrade.

7. MATLAB Script:

```
clc; clear; close all;

% Operating conditions
P_total = 30;          % Operating pressure in bar
W_span = [0 40000];    % Catalyst Weight Range (0 to 40,000 kg)
R_gas = 8.314;         % Universal Gas Constant in kJ/(kmol*K)

% Initial Conditions
F_P0 = 100;
F_N0 = 50;
F_A0 = 5;
F_G0 = 0;
F_H20 = 450;
T0 = 500 + 273.15;

Y0 = [F_P0; F_N0; F_A0; F_G0; F_H20; T0];

% Solve Mass and Energy Balance ODEs using Stiff Solver
[W, Y] = ode15s(@(W, Y) reforming_odes(W, Y, P_total, R_gas), W_span, Y0);

% Extract Integrated Profiles
F_P  = Y(:,1);
F_N  = Y(:,2);
F_A  = Y(:,3);
F_G  = Y(:,4);
F_H2 = Y(:,5);
T_C  = Y(:,6) - 273.15; % Convert Kelvin profile back to Celsius

% Conversion & Yield Calculations
X_N = (F_N0 - F_N) / F_N0;
Yield_A = (F_A - F_A0) / F_N0;

% PLOTTING RESULT
figure;

% Subplot 1: Composition Profile
subplot(3,1,1)
plot(W, F_P, '-r', 'LineWidth', 2); hold on;
plot(W, F_N, '-g', 'LineWidth', 2);
plot(W, F_A, '-b', 'LineWidth', 2);
plot(W, F_G, '-k', 'LineWidth', 2);
grid on;
ylabel('Flow Rate (kmol/hr)');
legend('Paraffins (P)', 'Naphthenes (N)', 'Aromatics (A)', 'Gases (G)', 'Location', 'best');
title('Composition Profile along the Reactor');
```

```
% Subplot 2: Temperature Profile
subplot(3,1,2)
plot(W, T_C, '-m', 'LineWidth', 2);
grid on;
ylabel('Temperature (\circC)');
title('Temperature Drop Profile (Adiabatic PFR)');

% Subplot 3: Conversion & Yield Profile
subplot(3,1,3)
plot(W, X_N, '--b', 'LineWidth', 2); hold on;
plot(W, Yield_A, '-m', 'LineWidth', 2);
grid on;
xlabel('Catalyst Weight (kg)');
ylabel('Fraction (0 to 1)');
legend('Conversion of Naphthenes', 'Yield of Aromatics', 'Location', 'best');
title('Conversion & Yield Profile');

% GOVERNING ODES FUNCTION
function dYdW = reforming_odes(W, Y, P_total, R_gas)
    % Y vector structure: [F_P, F_N, F_A, F_G, F_H2, T]
    F_P  = Y(1);
    F_N  = Y(2);
    F_A  = Y(3);
    F_G  = Y(4);
    F_H2 = Y(5);
    T    = Y(6); % Local Temperature in Kelvin

    F_total = sum(Y(1:5));

    % Calculate Component Partial Pressures (bar)
    P_P  = (F_P / F_total) * P_total;
    P_N  = (F_N / F_total) * P_total;
    P_A  = (F_A / F_total) * P_total;
    P_H2 = (F_H2 / F_total) * P_total;

    % Arrhenius Kinetic Rate Constants
    k1 = 4.5e4 * exp(-90000 / (R_gas * T));      % De-hydrogenation
    k2 = 1.2e3 * exp(-80000 / (R_gas * T));      % Isomerization
    k3 = 1.5e2 * exp(-95000 / (R_gas * T));      % Hydro cracking
    K_ep1 = 1e5;                                 % Thermodynamic Equilibrium Constant
```

```
    % Reaction rates
    r1 = k1 * (P_N - (P_A * P_H2^3) / K_ep1);
    r2 = k2 * (P_P - P_N);
    r3 = k3 * P_P * P_H2;

    % Enthalpies  of Reactions (kJ/kmol)
    dH1 = 210000;  % Highly Endothermic
    dH2 = 20000;   % Mildly Endothermic
    dH3 = -45000;  % Exothermic Hydro cracking

    % Pure Component Heat Capacities (kJ/kmol*K)
    Cp_P  = 250;
    Cp_N  = 220;
    Cp_A  = 180;
    Cp_G  = 80;
    Cp_H2 = 30;

    % Mass Balances
    dF_P_dW  = -r2 - r3;
    dF_N_dW  = r2 - r1;
    dF_A_dW  = r1;
    dF_G_dW  = 2 * r3;
    dF_H2_dW = 3 * r1 - r3;

    % Energy Balance (dT/dW)
    Numerator = (-dH1 * r1) + (-dH2 * r2) + (-dH3 * r3);
    Denominator = (F_P * Cp_P) + (F_N * Cp_N) + (F_A * Cp_A) + (F_G * Cp_G) + (F_H2 * Cp_H2);

    dT_dW = Numerator / Denominator;

    % Return Column Vector of Derivatives
    dYdW = [dF_P_dW; dF_N_dW; dF_A_dW; dF_G_dW; dF_H2_dW; dT_dW];
end
```

The middle temperature plot illustrates a steep drop from the inlet temperature of 500° C down to 300° C. This behavior highlights the extreme thermodynamic demand of the dehydrogenation reaction, which strips sensible heat out of the flowing stream. As the temperature crosses below 350° C, the system undergoes **kinetic freezing**. Because the kinetic rate constants depend exponentially on temperature via the Arrhenius equation, the reaction velocities drop to near-zero values, rendering the remaining 25,000 Kg of catalyst bed practically unutilized as indicated by the flattening of the concentration curves.

The bottom graph tracks the fractional parameters. A crucial feature observed is that the net yield of Aromatics climbs to ~ 90%, exceeding the standalone fractional conversion profile of the inlet naphthenes. This represents a vital industrial mechanism: as naphthenes are depleted, the ring-closure isomerization step continuously converts paraffins into fresh naphthenes, which instantly dehydrogenate into additional aromatics.

In conclusion, this **MATLAB** code provides a simplified yet insightful dynamic model of the catalytic reforming process, capturing key reaction pathways, thermal behavior, and catalyst performance in a plug flow reactor. It serves as a strong foundation for more detailed reactor design, process control, or optimization studies.

8. Conclusion:

A mathematical PFR simulation for catalytic naphtha reforming was successfully executed in MATLAB. The simulation highlights a major operating challenge: the highly endothermic nature of reforming reactions causes a rapid drop in reactor temperature, which severely limits high single-pass conversions due to thermal freezing.

To overcome this limitation on an industrial scale, it is concluded that a single large PFR bed is inefficient. Refineries must deploy a multi-bed reactor design (typically 3 to 4 adiabatic reactors arranged in series) equipped with **Interstage Fired Heaters** to periodically reheat the stream back to 500° C ensuring optimal catalyst utilization and maximum high-octane gasoline yield.

9. References:

1. Fogler, H. S. Elements of Chemical Reaction Engineering, 5th Ed., Prentice Hall, 2016.
2. Gary, J. H., Handwerk, G. E., Kaiser, M. J. Petroleum Refining: Technology and Economics.
3. Rostrup-Nielsen, J. Catalytic Reforming: Science and Technology.